

Day 2 — The Magic Constant

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) derive the magic constant $S = n(n^2 + 1)/2$ for the 3×3 case by direct argument, and (b) apply the formula to predict the constant for $n = 4, 5, 6, 7$.

Materials

- Whiteboard
- Notebook + pen
- The 3×3 magic square from Day 1, visible on the board

Vocabulary review

- *magic square, magic constant* — defined Day 1.

Lesson flow

Warm-up (5 min)

- Recall chant: “ 3×3 square with 1 to 9 — magic constant is ____?” Class: “15.”
- “Why exactly 15, though? Why not 14 or 17?” Take 2 student responses.

Core (20 min)

1. **The sum argument.** Write on the board:

Add up all the numbers in the square: $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45$.

“This total goes into the square once. Now — the square has 3 rows. Each row has the same sum S . So $3 \times S = 45 \rightarrow S = 15$.”

Repeat the logic with student input. They should arrive at the formula step-by-step.

2. **Generalise.** For an $n \times n$ square containing the numbers 1, 2, ..., n^2 :

Sum of all numbers = $1 + 2 + \dots + n^2 = n^2(n^2 + 1) / 2$

The square has n rows, each summing to S . So:

$n \times S = n^2(n^2 + 1) / 2 \rightarrow S = n(n^2 + 1) / 2$

3. **Apply.** Compute together for several n :

n	n^2	n^2+1	$n(n^2+1)/2$
3	9	10	$3 \times 10 / 2 = \mathbf{15}$
4	16	17	$4 \times 17 / 2 = \mathbf{34}$
5	25	26	$5 \times 26 / 2 = \mathbf{65}$

n	n^2	n^2+1	$n(n^2+1)/2$
6	36	37	$6 \times 37 / 2 = \mathbf{111}$
7	49	50	$7 \times 50 / 2 = \mathbf{175}$

4. **A paraphrase from the source.** Nārāyaṇa Paṇḍita, in *Bhadragaṇita* (Book 14 of *Gaṇita-kaumudī*, 1356 CE), gives the magic-constant rule explicitly: the common sum is one n -th of the sum of the numbers in the square. We’ve just re-derived his rule. (See *sources.md*.)

Activity (15 min) — Formative quiz Part A

- Hand out quizzes/formative.md Part A.
- Students complete individually. 10 minutes; 3 minutes for self-check against the board answer key.
- Part A is 6 short questions: 3 on the formula, 3 on verifying given squares.

Wrap-up (5 min)

- “Quick check: what would the magic constant be for a 10×10 magic square with the numbers 1 to 100?” Class works it out: $S = 10 \times 101 / 2 = \mathbf{505}$.
- Preview Day 3: “Tomorrow we ask: how many different 3×3 magic squares are there with 1 to 9? Take guesses tonight.”

Student-facing content

The magic constant formula.

For an $n \times n$ magic square containing the numbers 1, 2, ..., n^2 :

$$S = n(n^2 + 1) / 2$$

Why? The sum of all n^2 numbers is $n^2(n^2 + 1) / 2$ (the standard sum-to- n formula). This total is split equally among the n rows, each of which sums to S . So $n \times S = n^2(n^2 + 1) / 2 \rightarrow$ divide both sides by n .

n	Constant
3	15
4	34
5	65
6	111
7	175

Nārāyaṇa Paṇḍita stated this rule in *Bhadragaṇita* (1356 CE).

Homework

- Verify the formula for $n = 8$ and $n = 9$. (Answers: 260 and 369.)
- For each, also compute the total sum $n^2(n^2 + 1)/2$ directly and confirm it equals $n \times S$.

Differentiation

- **One tier up:** What if the magic square contains *consecutive* numbers but starting at k , not at 1? (E.g. a 3×3 with 5, 6, 7, ..., 13.) Derive a formula for the new magic constant. (Answer: $S = n \times k + n(n^2-1)/2$.)
- **One tier down:** Focus on the 3×3 case alone. Verify the constant 15 by summing every row, column, and diagonal of the Day-1 square. Do it slowly.

Teacher notes

- A common stumble: students compute $1 + 2 + \dots + 9$ incorrectly. Remind them that $1+9$, $2+8$, $3+7$, $4+6$ each = 10, plus 5 = 45. This itself is a pretty argument worth pausing on.
- For the generalisation, some students will struggle with the n^2 notation. Walk them through the $n = 5$ case explicitly: $1 + 2 + \dots + 25 = 25 \times 26 / 2 = 325$; split among 5 rows $\rightarrow 65$.
- The formula $S = n(n^2 + 1)/2$ is **the** rule of this module. Drill it.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — paraphrased here: the magic constant equals one n -th of the total sum of the n^2 entries.